Project Title: Retail Sales and Profitability Analysis Using Power BI

Project Objective: To analyze retail superstore sales data and build an interactive Power BI dashboard to monitor business performance, track sales vs target, identify top-performing products and states, understand seasonality, profit leakage, and provide data-driven recommendations.

Dataset Description: Superstore Sales Data with 500+ orders containing sales, profit, category, sub-category, geographical and customer details. Spans multiple years with State, City, Category (Electronics, Furniture, Clothing), Sub-Category (Printers, Bookcases, Saree, Phones etc.), Sales, Quantity, Discount, Profit.

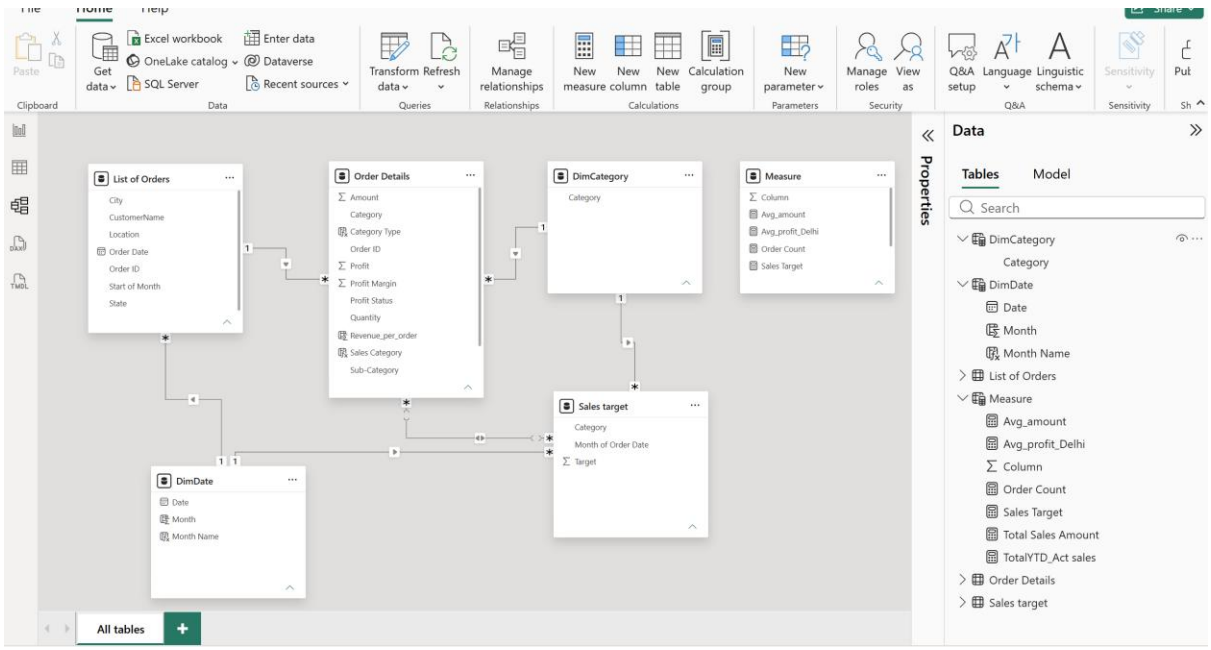
Column Description:

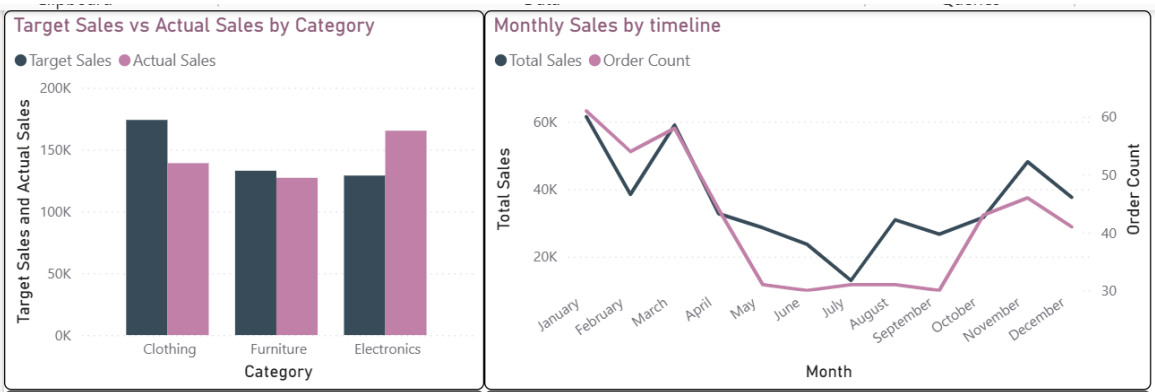
Column Name	Description
Order ID	Unique identifier for each order
Order Date	Date of order - used for monthly trend analysis (Peak Feb-Mar)
Customer	Customer name and segment type
City / State	Location - MP 101 & Maharashtra 90 highest orders, TN 8 lowest
Category	Main category - Electronics, Furniture, Clothing
Sub-Category	Product detail - Printers 58K, Bookcases 57K, Saree 54K, Phones 46K top sellers (60%+ sales)
Sales	Revenue amount - Total 431.5K vs Target 435.9K
Quantity	Units sold - high quantity shows low profit due to discounts
Discount	Discount percentage - main cause of profit leakage
Profit	Profit amount - negative for bulk orders
Profit Margin	Calculated as Profit/Sales - for donut chart
Sales Target	Target value 435.9K for KPI comparison

Data cleaning & transformation:

Converted Order Date to date, created DAX Measures (Total Sales, Profit Margin, Total Orders), standardized state names, handled negative profit and >80% discount outliers.

Visualization and Insights Arrived with screenshot of visualization





₹ 431.502K
Total Sales Amount

₹ 435.9K
Sales Target

Clothing

12K
Min of Target

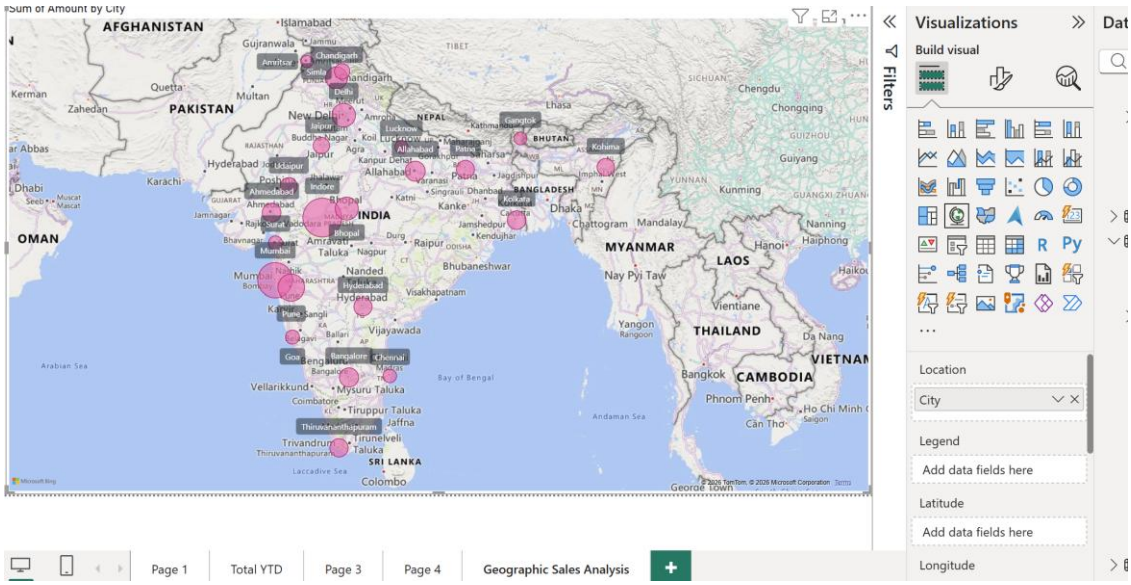
Electronics

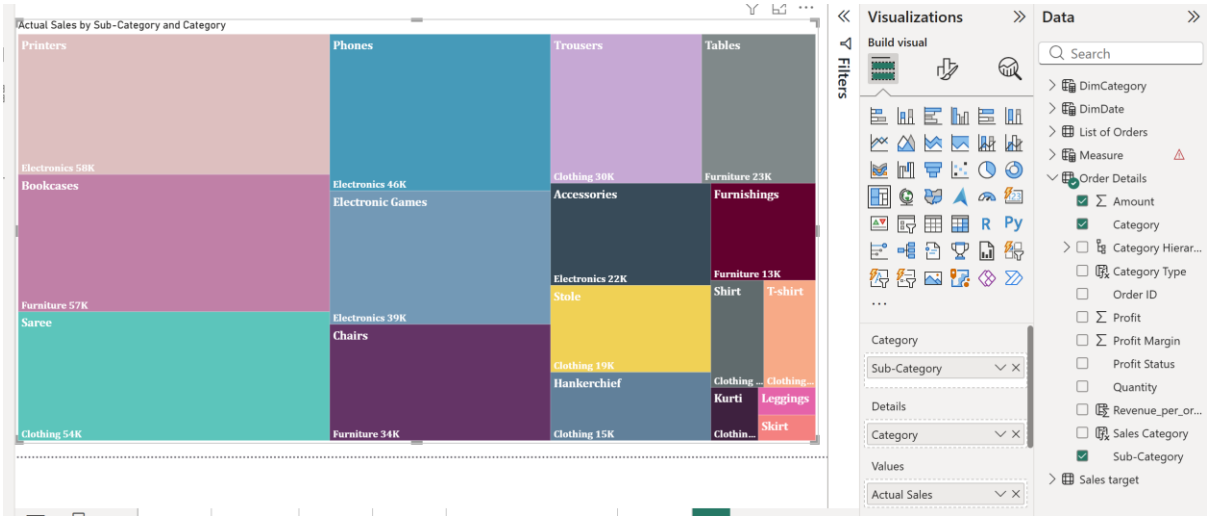
9K
Min of Target

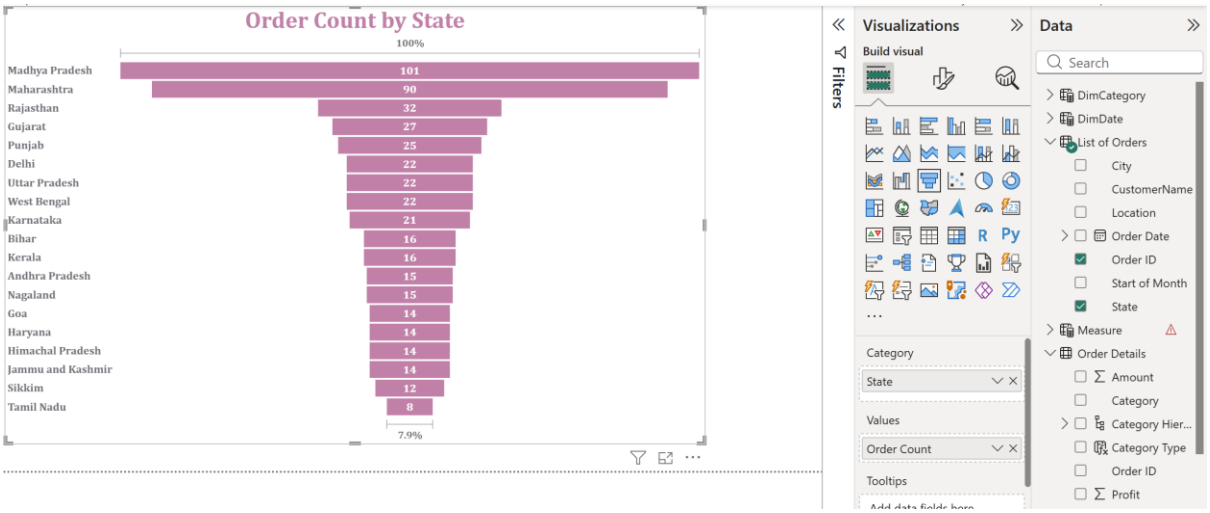
Furniture

10K
Min of Target

Category	Actual Amount	Sales Target
⊟ Clothing	139054	174000
⊟ 2018	94601	126000
April	13478	12000
May	9518	12000
June	8782	12000
July	2981	14000
August	11822	14000
September	10717	14000
October	11488	16000
November	16270	16000
December	9545	16000
⊟ 2019	44453	48000
January	13466	16000
February	9569	16000
March	21418	16000
⊟ Electronics	165267	129000
⊞ 2018	105098	81000
⊞ 2019	60169	48000
⊟ Furniture	127181	132900
⊞ 2018	73003	98000
⊞ 2019	54178	34900
Total	431502	435900







Report/Dashboard screenshot:

1. Descriptive Analysis – What happened?

- Achieved 99% of the sales target.
- MP and Maharashtra contributed 38% of total orders.
- Four sub-categories accounted for more than 60% of total sales.

2. Diagnostic Analysis – Why did it happen?

- Clothing sales underperformed by ₹35K.
- High discounts on bulk orders resulted in profit loss.
- Limited marketing activities in South and Northeast India may have contributed to lower sales in these regions.

3. Predictive Analysis – What could happen next?

- Heavy dependence on MP and Maharashtra could create a business risk.
- Untapped markets in South India may limit future growth.
- Continuing high discounts on bulk orders could further reduce profitability.

- If the historical seasonal pattern repeats, sales during Q2 and Q3 could decline significantly.

4. Prescriptive Analysis – What actions should be taken?

- Defend: Maintain strong sales performance in MP and Maharashtra.
- Expand: Focus on growing sales in Tamil Nadu, Karnataka, and Sikkim.
- Prioritize: Invest in the four best-performing sub-categories to maximize revenue.
- Improve profitability: Set discount limits and promote products such as Stoles and Handkerchiefs to improve profit margins.
- Reduce seasonality: Introduce targeted offers and promotional campaigns during Q2 and Q3 to improve sales consistency.